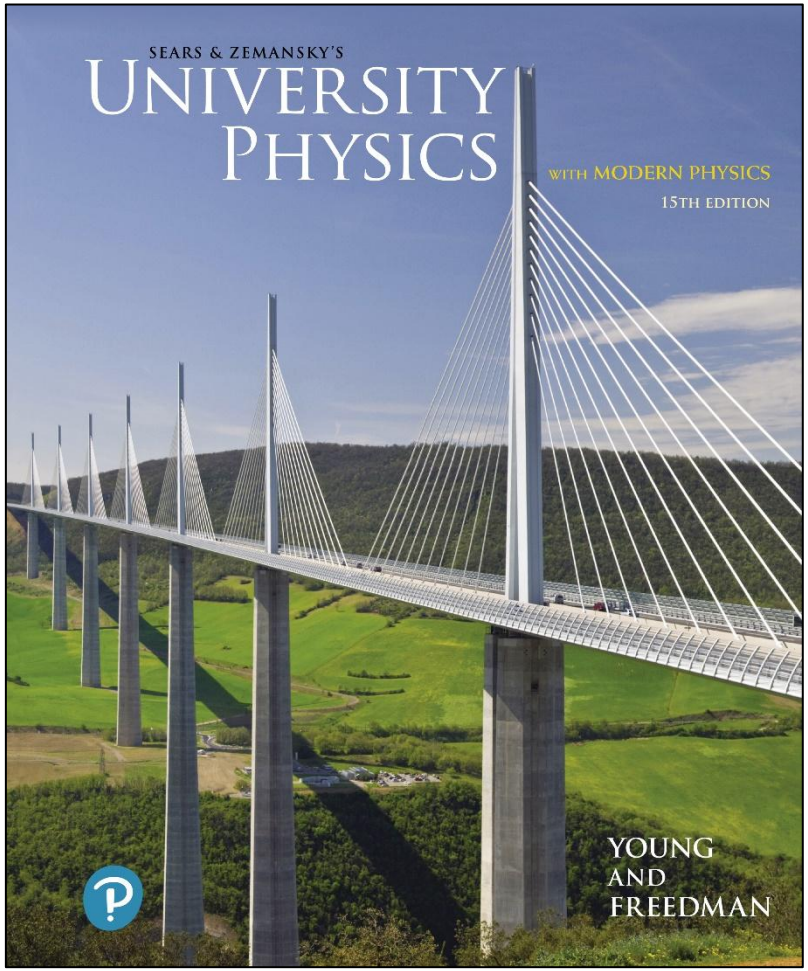


Chapter 14

Periodic Motion

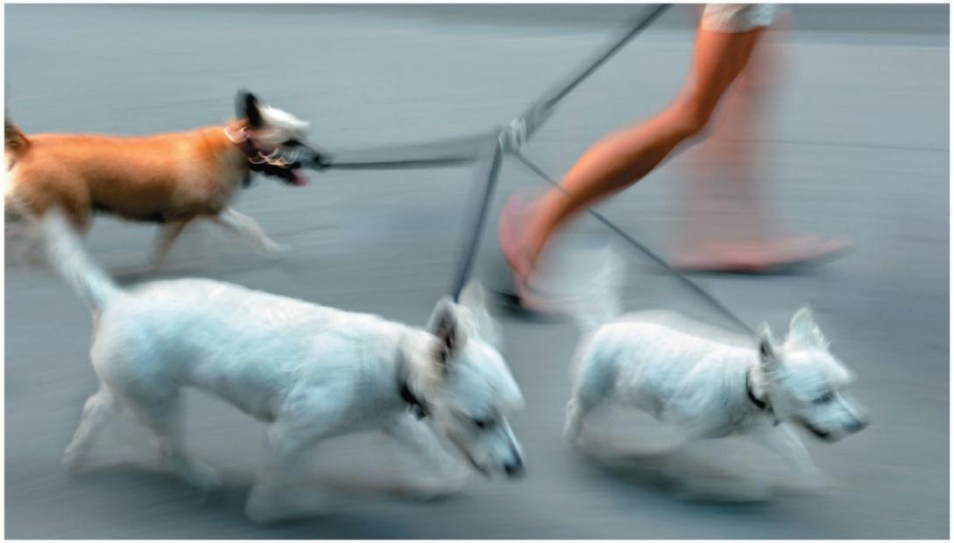
Learning Outcomes

In this chapter, you'll learn…

- how to describe oscillations in terms of amplitude, period, frequency, and angular frequency.
- how to apply the ideas of **simple harmonic motion** to different physical situations.
- how to analyze the motions of a pendulum.
- what determines how rapidly an oscillation dies out.
- how a driving force applied to an oscillator at a particular frequency can cause a very large response, or **resonance**.

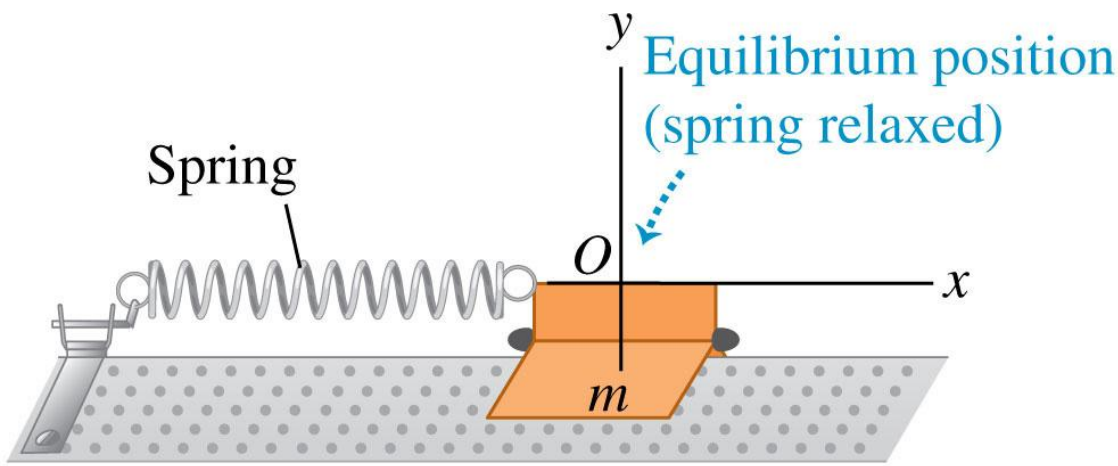
Introduction

- Why do dogs walk faster than humans? Does it have anything to do with the characteristics of their legs?
- Many kinds of motion (such as a pendulum, musical vibrations, and pistons in car engines) repeat themselves. We call such behavior **periodic motion** or **oscillation**.

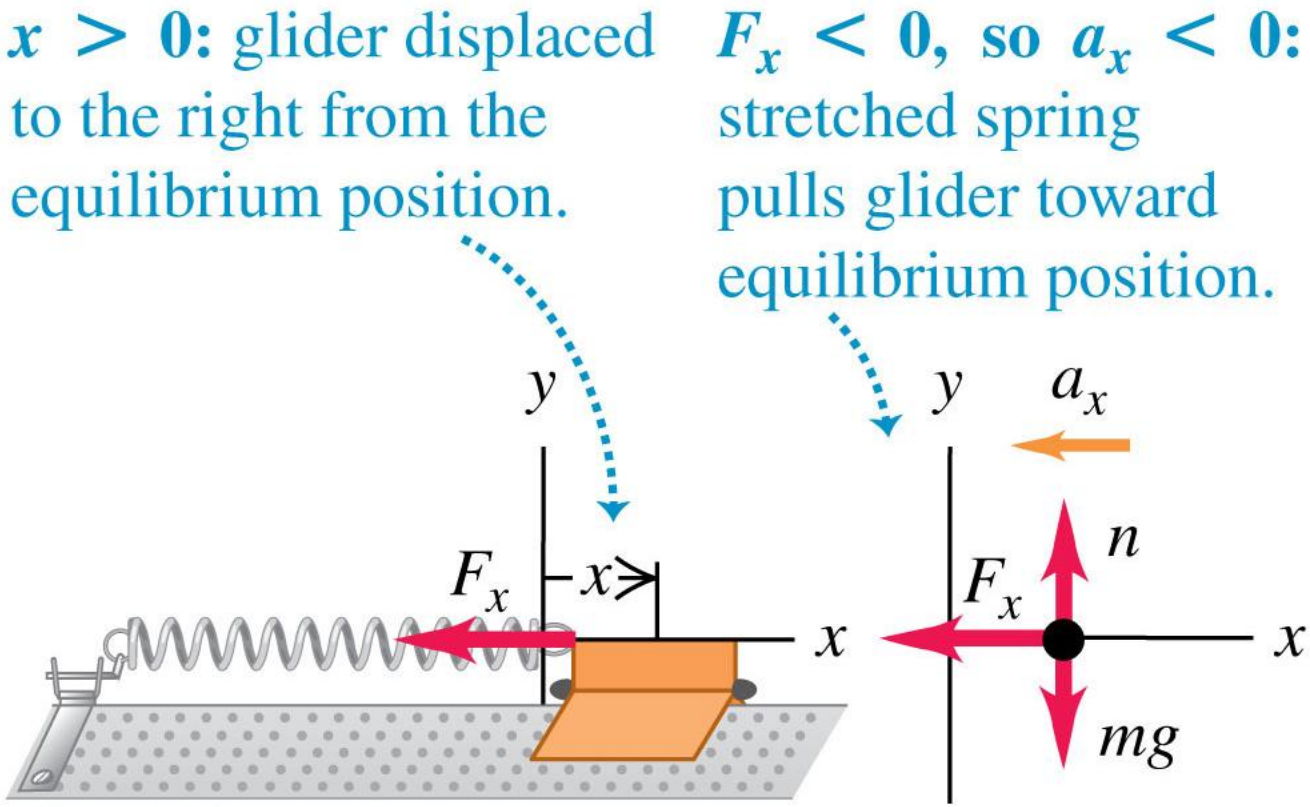


What Causes Periodic Motion? (1 of 4)

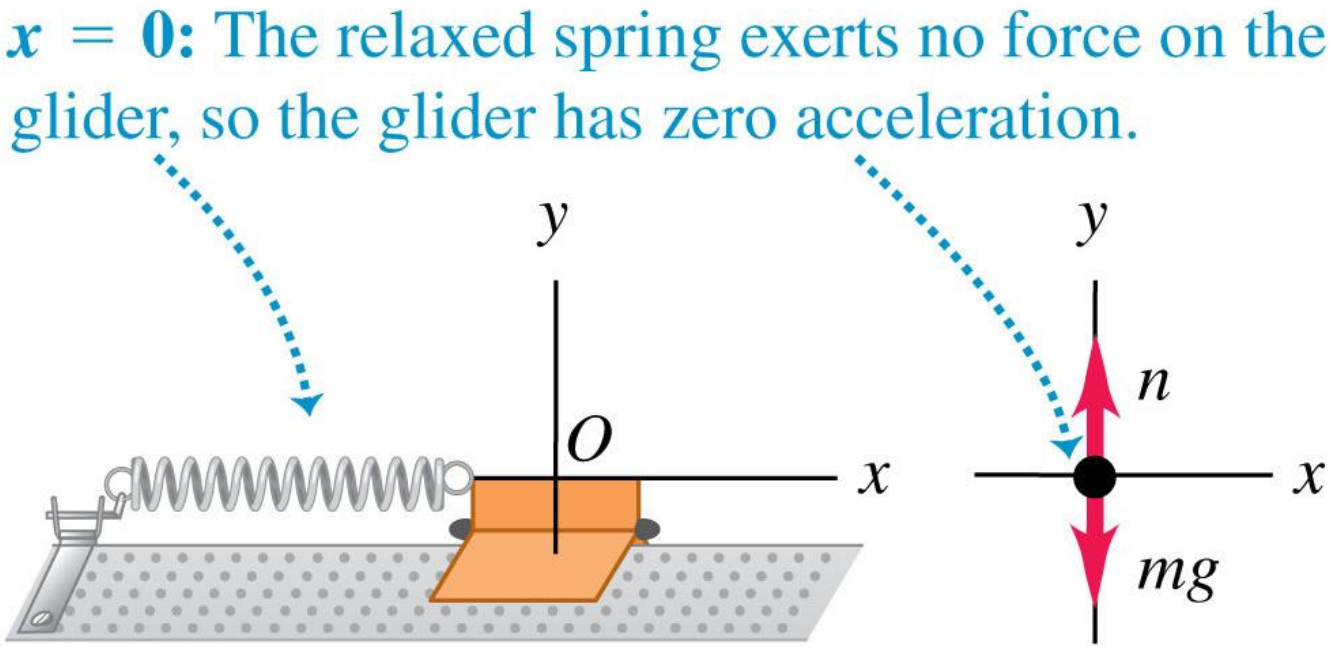
- If an object attached to a spring is displaced from its equilibrium position, the spring exerts a **restoring force** on it, which tends to restore the object to the equilibrium position.
- This force causes **oscillation** of the system, or **periodic motion**.



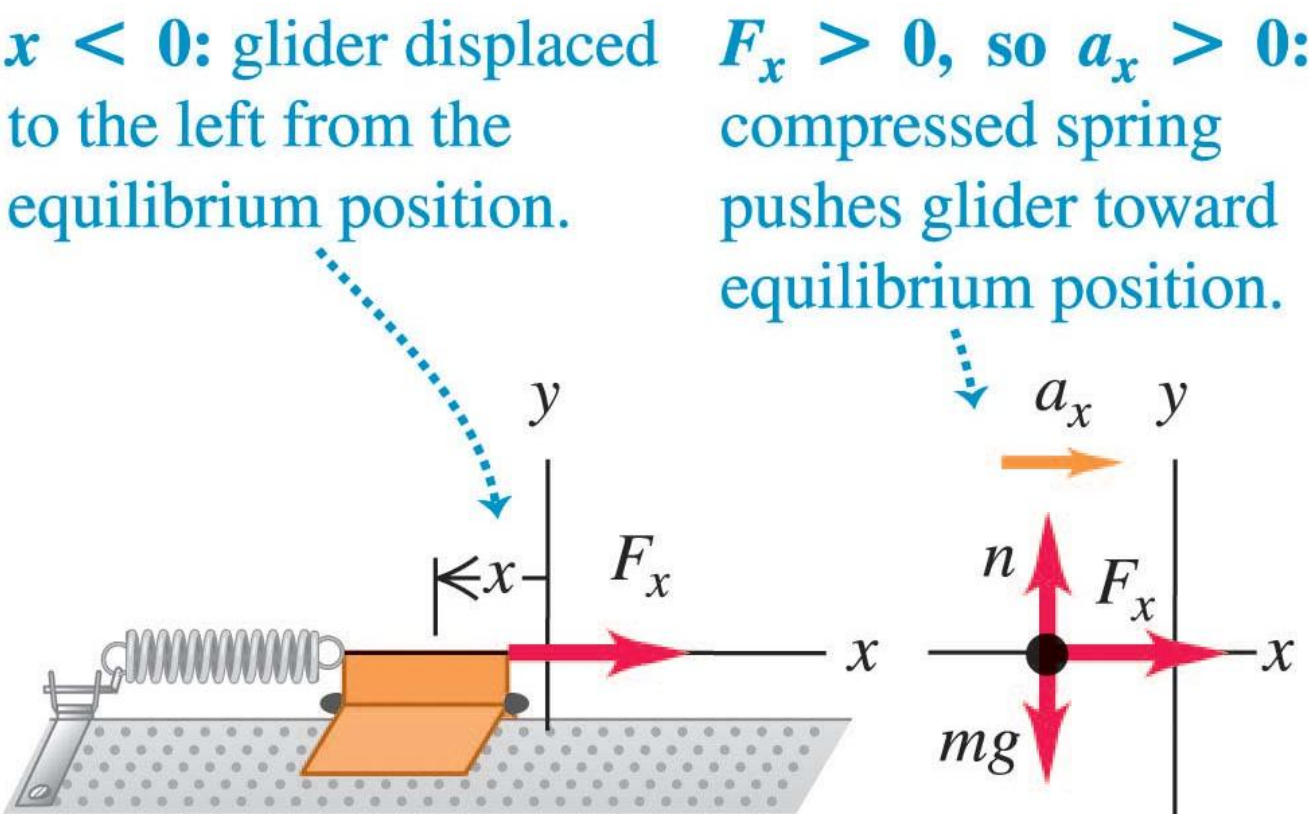
What Causes Periodic Motion? (2 of 4)



What Causes Periodic Motion? (3 of 4)



What Causes Periodic Motion? (4 of 4)

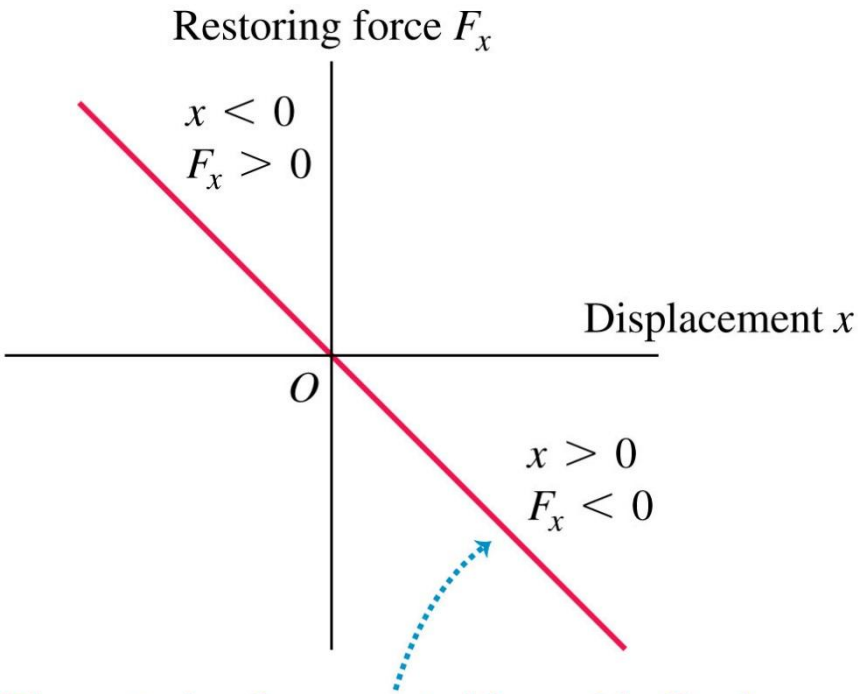


Characteristics of Periodic Motion

- The **amplitude**, *A*, is the maximum magnitude of displacement from equilibrium.
- The **period**, *T*, is the time for one cycle.
- The **frequency**, *f*, is the number of cycles per unit time.
- The **angular frequency**, *ω,* is 2π times the frequency: $\omega = 2\pi f$.
- The frequency and period are reciprocals of each other: $f = \frac{1}{T}$ and $T = \frac{1}{f}$.

Simple Harmonic Motion (SHM) (1 of 2)

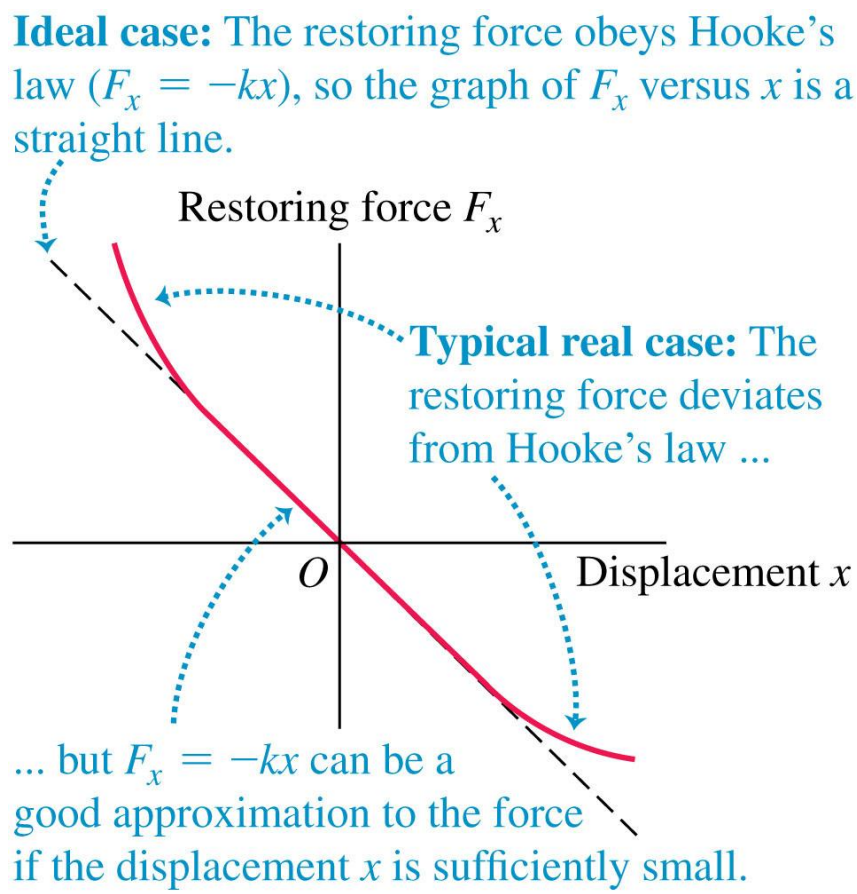
- When the restoring force is **directly proportional** to the displacement from equilibrium, the resulting motion is called **simple harmonic motion** (SHM).



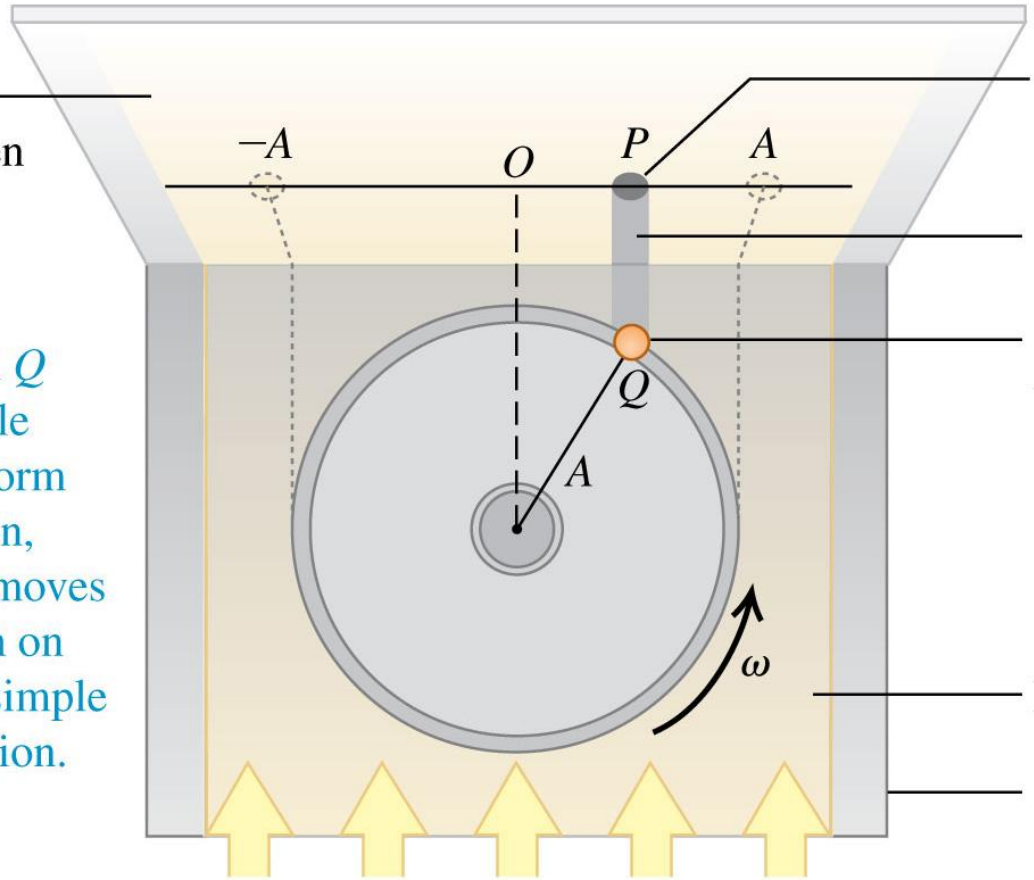
The restoring force exerted by an idealized spring is directly proportional to the displacement (Hooke's law, $F_x = -kx$): the graph of F_x versus x is a straight line.

Simple Harmonic Motion (SHM) (2 of 2)

- In many systems, the restoring force is approximately proportional to displacement if the displacement is sufficiently small.
- That is, if the amplitude is small enough, the oscillations are approximately simple harmonic.



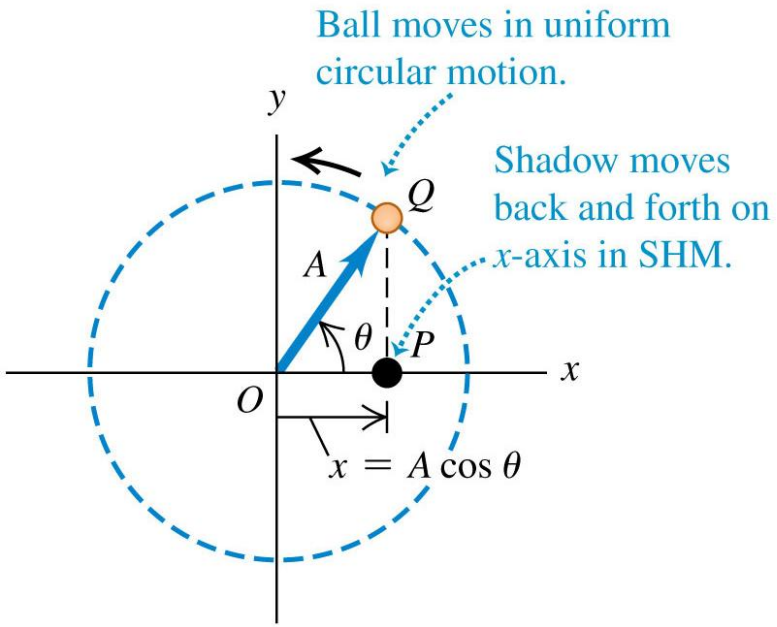
Simple Harmonic Motion Viewed as a Projection (1 of 2)



While the ball Q on the turntable moves in uniform circular motion, its *shadow* P moves back and forth on the screen in simple harmonic motion.

Simple Harmonic Motion Viewed as a Projection (2 of 2)

- The circle in which the ball moves so that its projection matches the motion of the oscillating object is called the **reference circle**.
- As point *Q* moves around the reference circle with constant angular speed, vector *OQ* rotates with the same angular speed.
- Such a rotating vector is called a **phasor**.



Characteristics of SHM (1 of 2)

- For an object of mass *m* vibrating by an ideal spring with a force constant *k*:

Angular frequency for simple harmonic motion

$$\omega = \sqrt{\frac{k}{m}}$$

k: Force constant of restoring force

m: Mass of object

Frequency for simple harmonic motion

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

ω: Angular frequency

k: Force constant of restoring force

m: Mass of object

Period for simple harmonic motion

$$T = \frac{1}{f} = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

f: Frequency

ω: Angular frequency

m: Mass of object

k: Force constant of restoring force

- Video Tutor Solution: Example 14.2

Suppose a horizontal block-spring system has a spring constant 1,145 N/m and block of mass 11 kg. Calculate the angular frequency in rad/s of its harmonic motion..

Suppose a horizontal block-spring system has a spring constant of 1240 N/m and a block of mass 15.0 kg. Calculate:

- the angular frequency in rad/s: [Select]
- the frequency in Hz: [Select], and
- the period in s: [Select].

Characteristics of SHM (2 of 2)

- The greater the mass *m* in a tuning fork's tines, the lower the frequency of oscillation, and the lower the pitch of the sound that the tuning fork produces.

Displacement as a Function of Time in SHM (1 of 5)

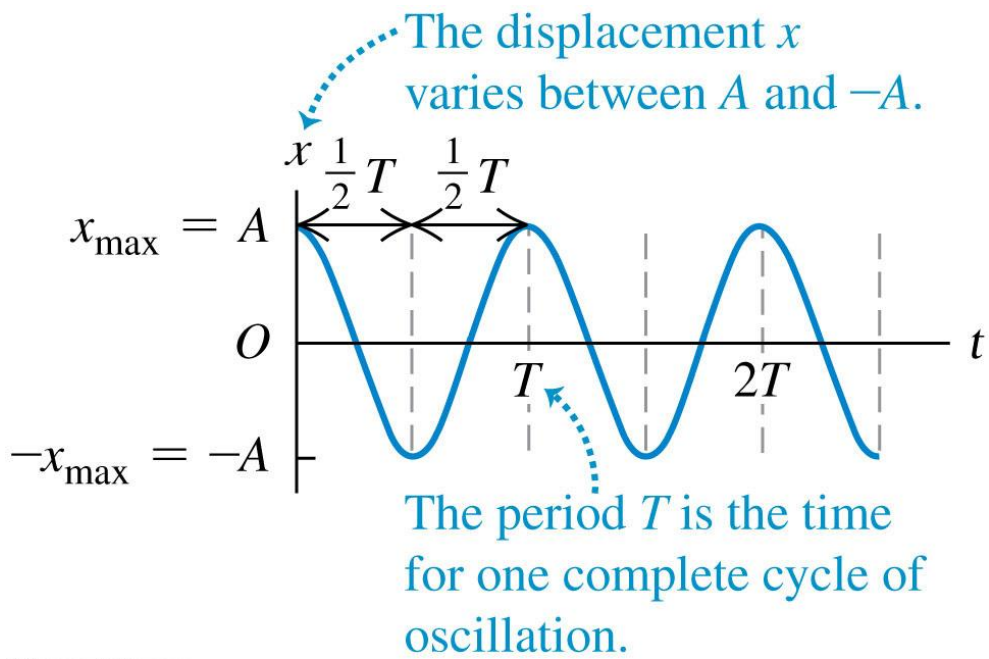
- The displacement as a function of time for SHM is:

Displacement in simple harmonic motion as a function of time

Amplitude, Time, Phase angle

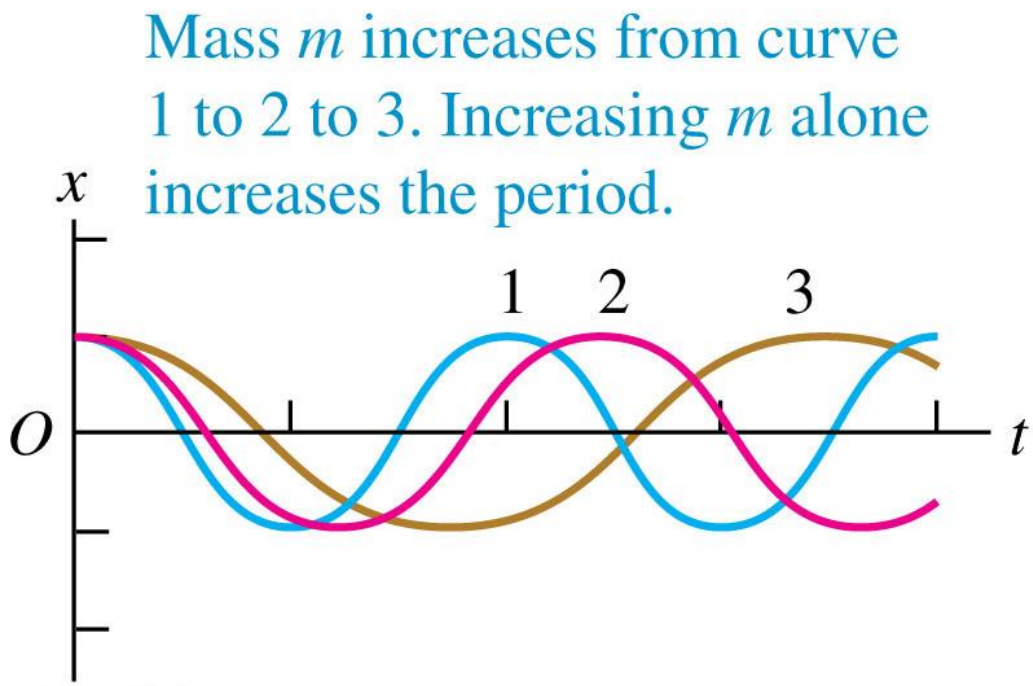
$$x = A\cos(\omega t + \phi)$$

Angular frequency $= \sqrt{k/m}$



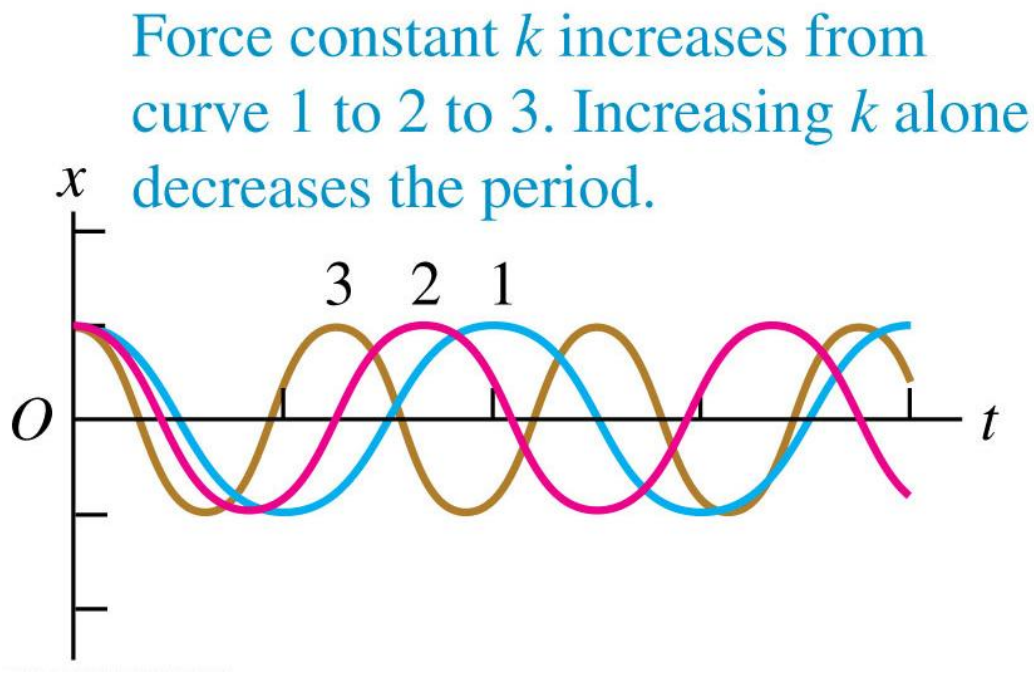
Displacement as a Function of Time in SHM (2 of 5)

- Increasing *m* with the same *A* and *k* increases the period of the displacement vs time graph.



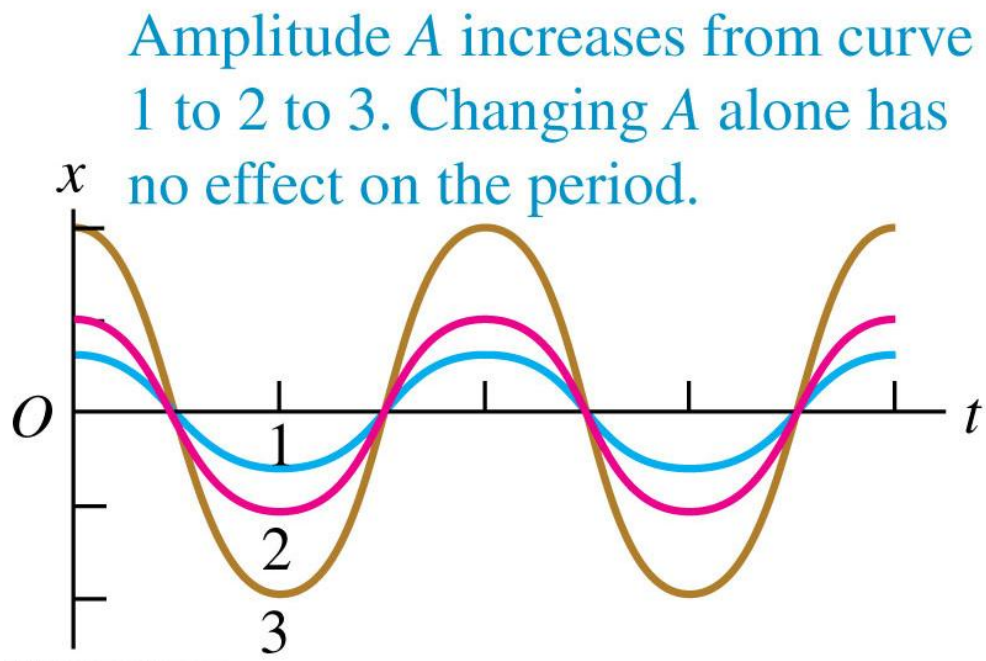
Displacement as a Function of Time in SHM (3 of 5)

- Increasing *k* with the same *A* and *m* decreases the period of the displacement vs time graph.



Displacement as a Function of Time in SHM (4 of 5)

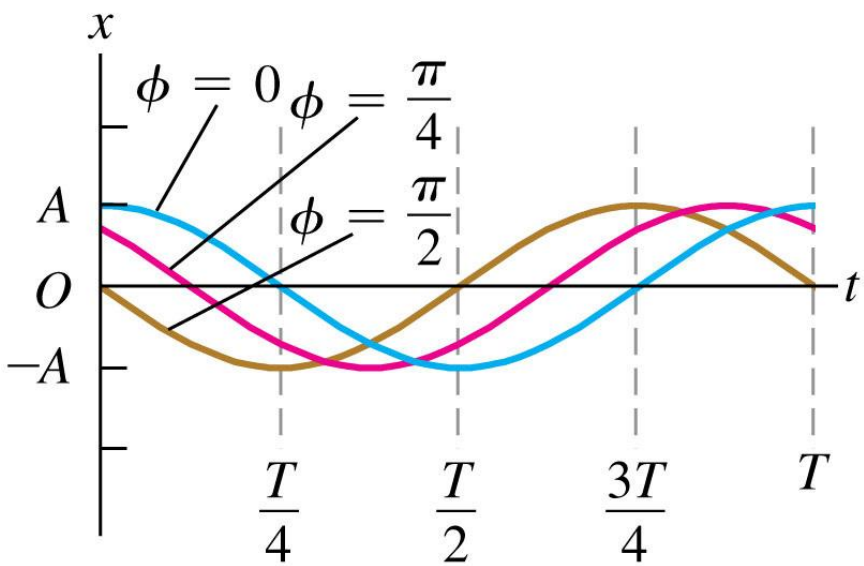
- Increasing *A* with the same *m* and *k* does not change the period of the displacement vs time graph.



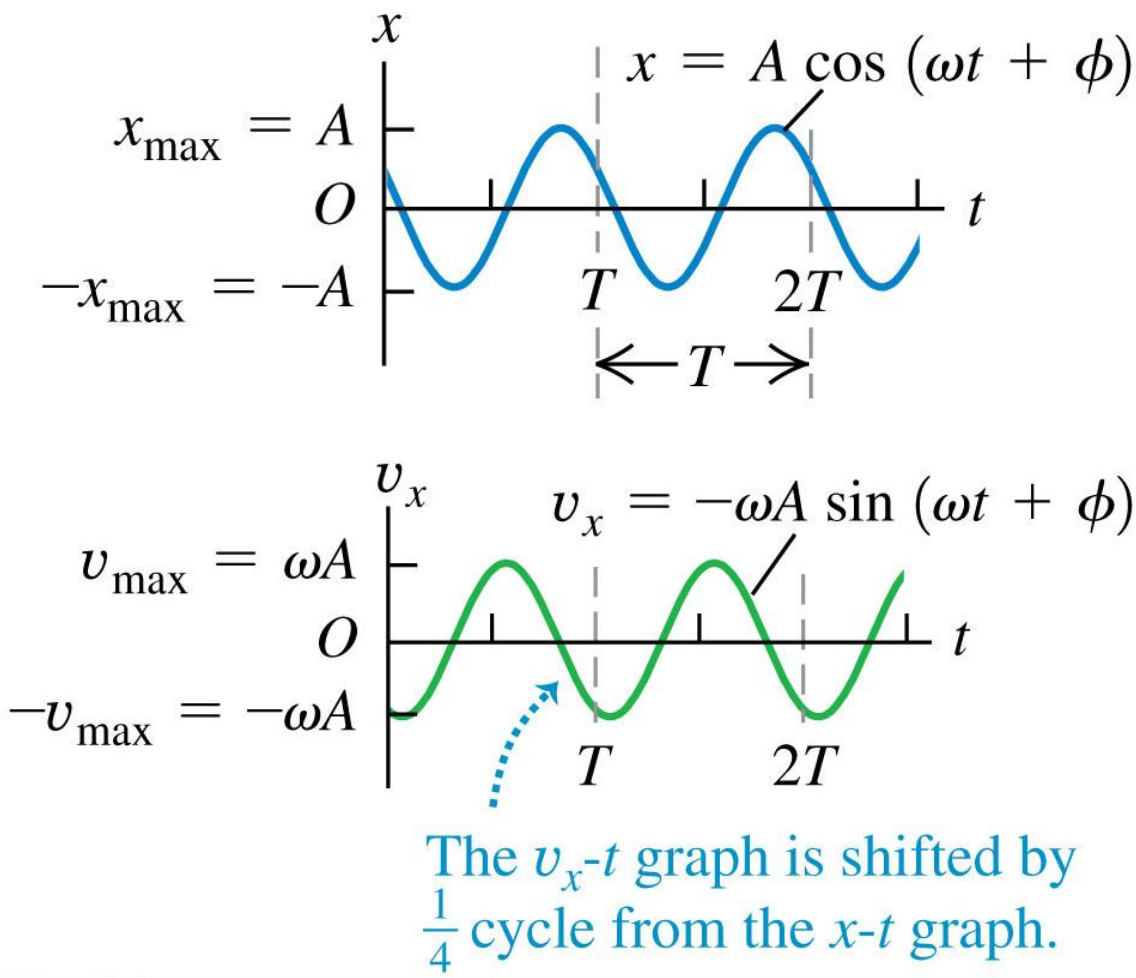
Displacement as a Function of Time in SHM (5 of 5)

- Increasing ϕ with the same A, m, and k only shifts the displacement vs time graph to the left.

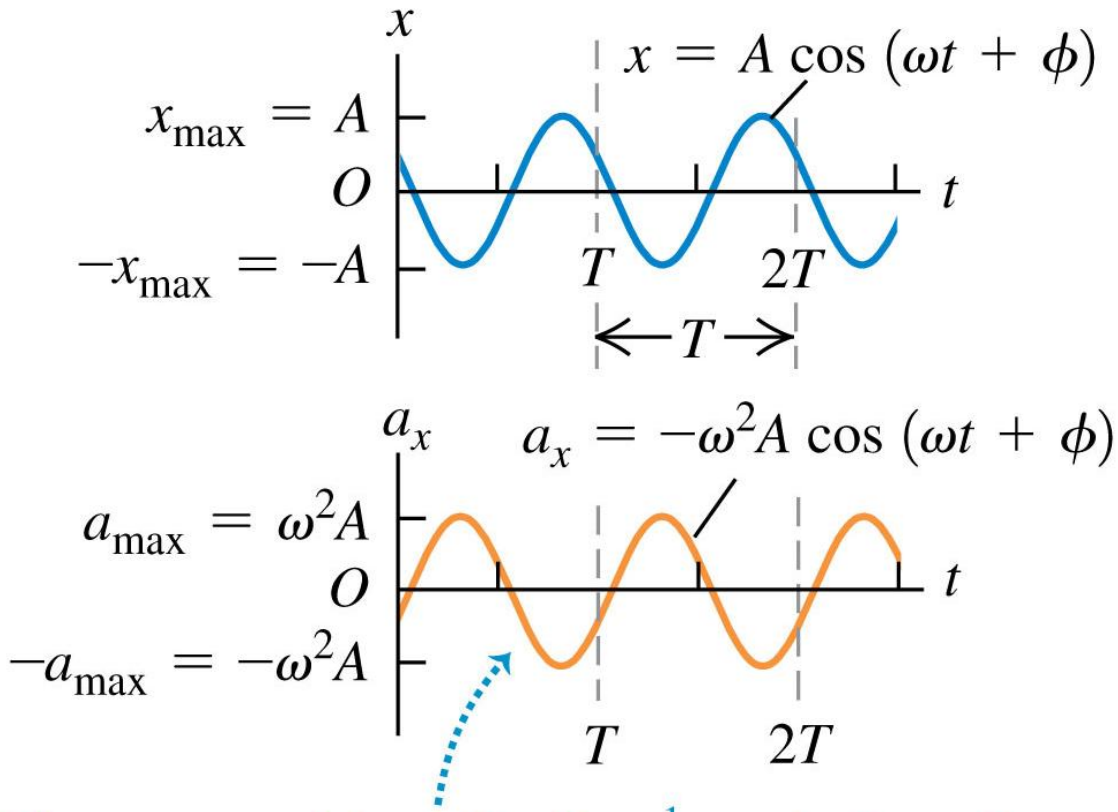
These three curves show SHM with the same period T and amplitude A but with different phase angles ϕ.



Graphs of Displacement and Velocity for SHM



Graphs of Displacement and Acceleration for SHM



The a_x-t graph is shifted by $\frac{1}{4}$ cycle from the v_x-t graph and by $\frac{1}{2}$ cycle from the x-t graph.

Fill In the Blanks with numerical values. ***Give three digits for each answer***, rounding as appropriate. For example, if your answer is "2", report "2.00". If your answer is "1.5555", report "1.56". If your answer is "0.7", report "0.700".

Suppose a mass-spring system has the description $x(t) = 4.00 \cos(2.00t - 0.500)$, where each number has unshown MKS units.

The amplitude is _____ m, the angular frequency is _____ rad/s, the phase shift is _____ rad, the frequency is _____ Hz; the period is _____ s.

Choose the correct answers.

Suppose a simple harmonic oscillator solution for a mass-spring system is given by $x(t) = 0.7 \cos(8t + 1.9)$, where all MKS units are assumed.

For this solution,

(a) the amplitude is [Select] m;

(b) the angular frequency is [Select] rad/s;

(c) the phase shift is [Select] rad;

(d) the frequency is [Select] Hz;

(e) the period is [Select] s;

(f) the maximum speed is [Select] m/s;

(g) the maximum acceleration is [Select] m/s^2;

(h) the position at t = 0 s is [Select] m;

(i) the velocity at t = 0 s is [Select] m/s;

(j) the acceleration at t = 0 s is [Select] m/s^2;

(k) Find the first time t > 0 that the mass reaches x = 0. [Select]

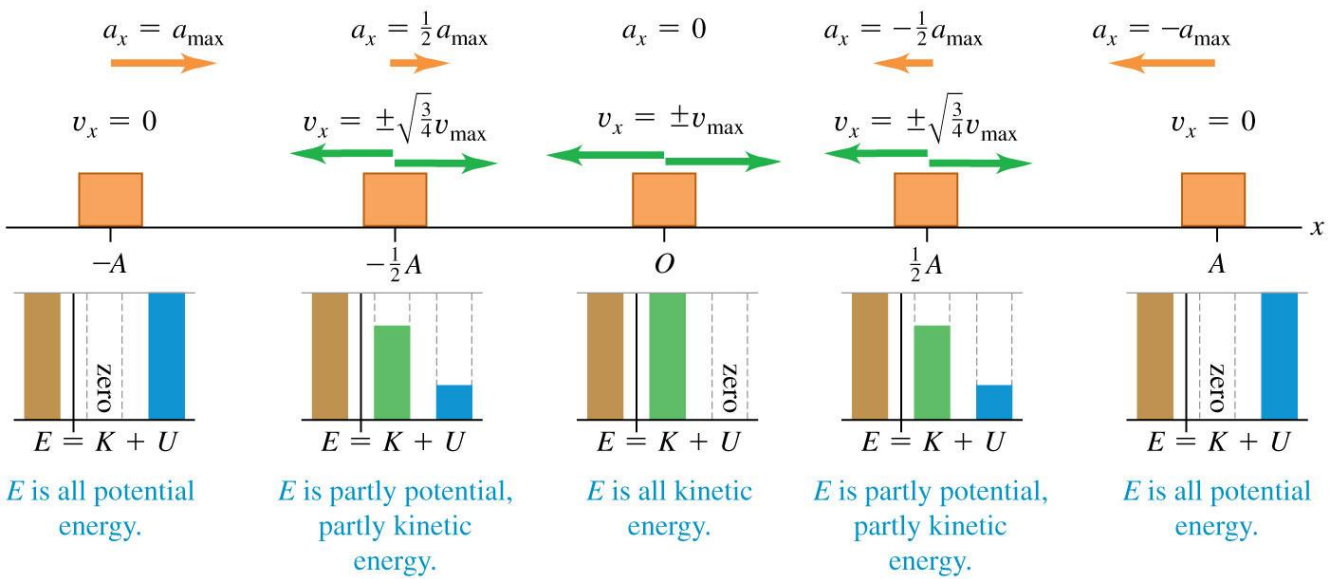
(l) Find the first time t > 0 that the speed is 0. [Select]

Suppose a mass-spring system has spring constant 755 N/m and mass 6.4 kg. The mass is displaced in the negative x-direction by 0.21 m and set in motion at 2.13 m/s in the positive x-direction. Calculate the phase shift as a positive angle in radians, assuming a cosine solution.

Energy in SHM

- The total mechanical energy $E = K + U$ is conserved in SHM:

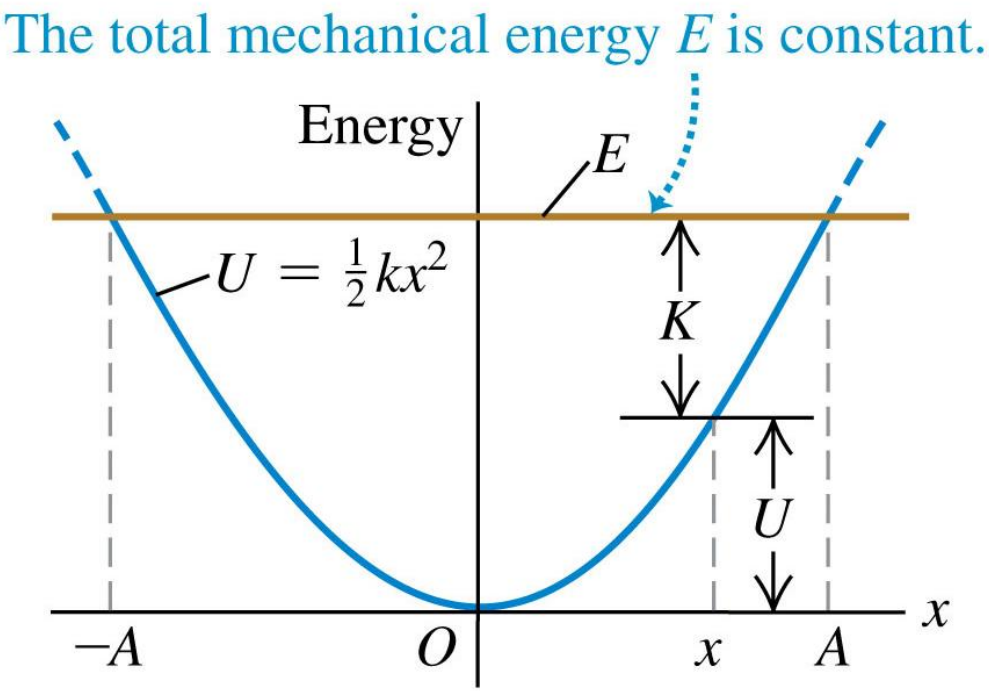
$$E = \frac{1}{2}mv_x^{\ 2} + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 = \text{constant}$$



- Video Tutor Solution: Example 14.4

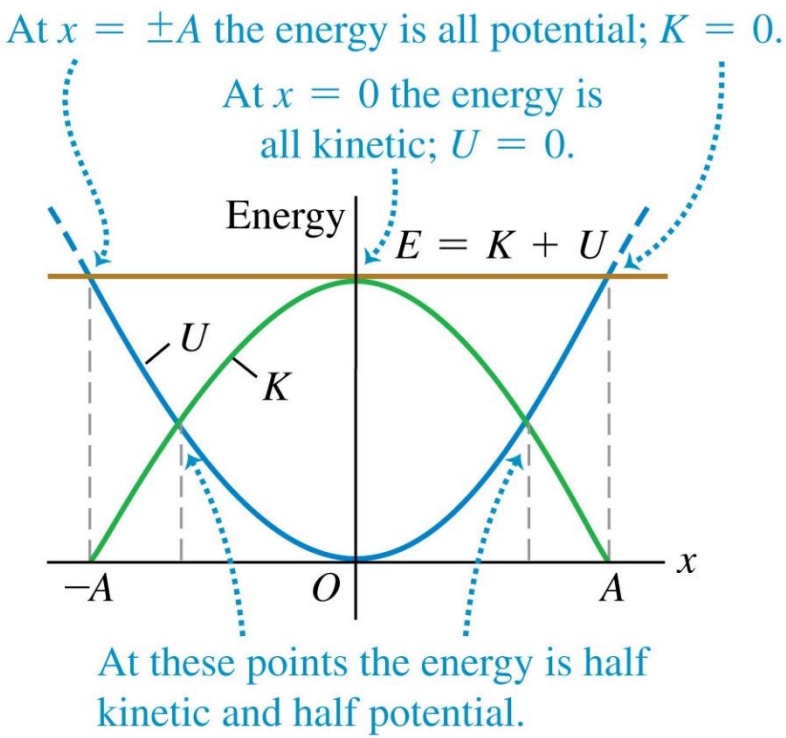
Energy Diagrams for SHM (1 of 2)

- The potential energy *U* and total mechanical energy *E* for an object in SHM as a function of displacement *x*.



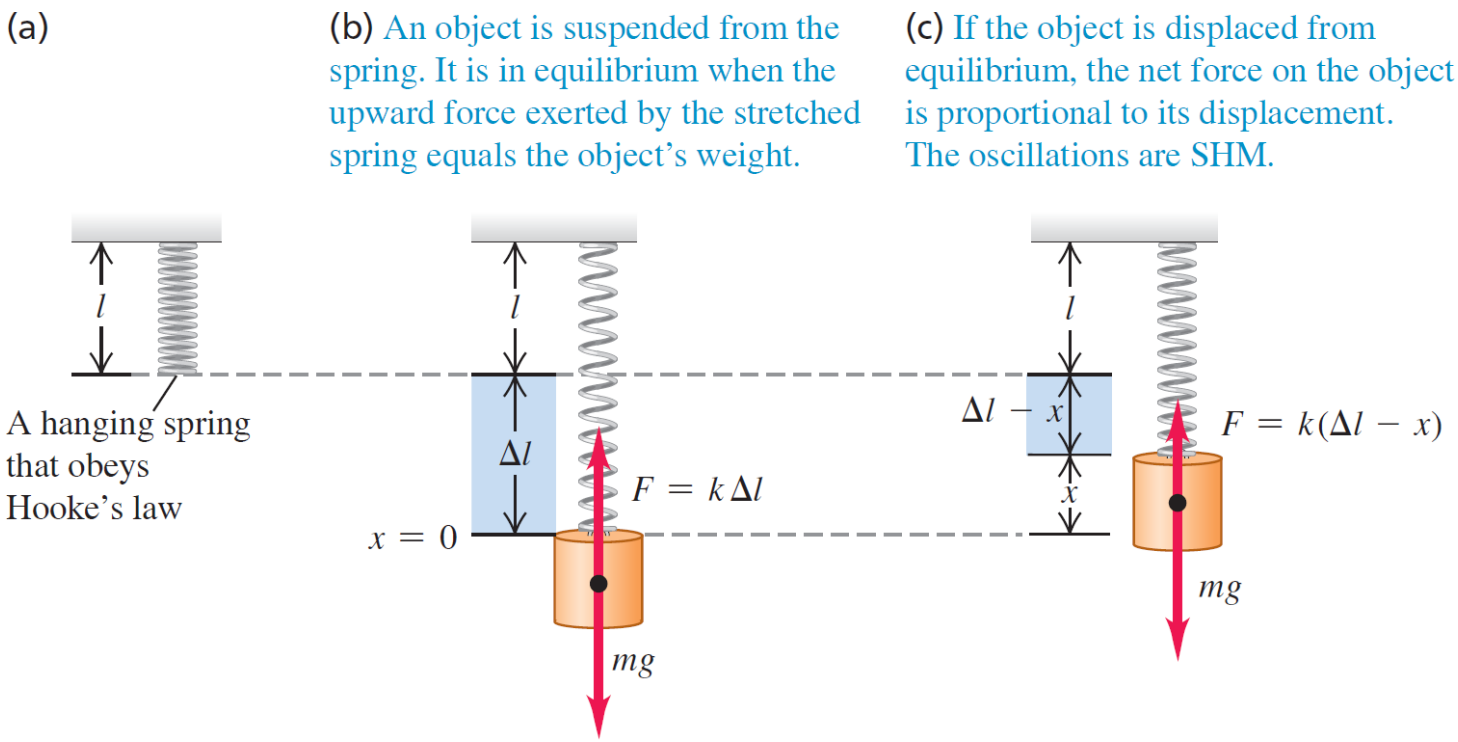
Energy Diagrams for SHM (2 of 2)

- The potential energy *U*, kinetic energy *K*, and total mechanical energy *E* for an object in SHM as a function of displacement *x*.



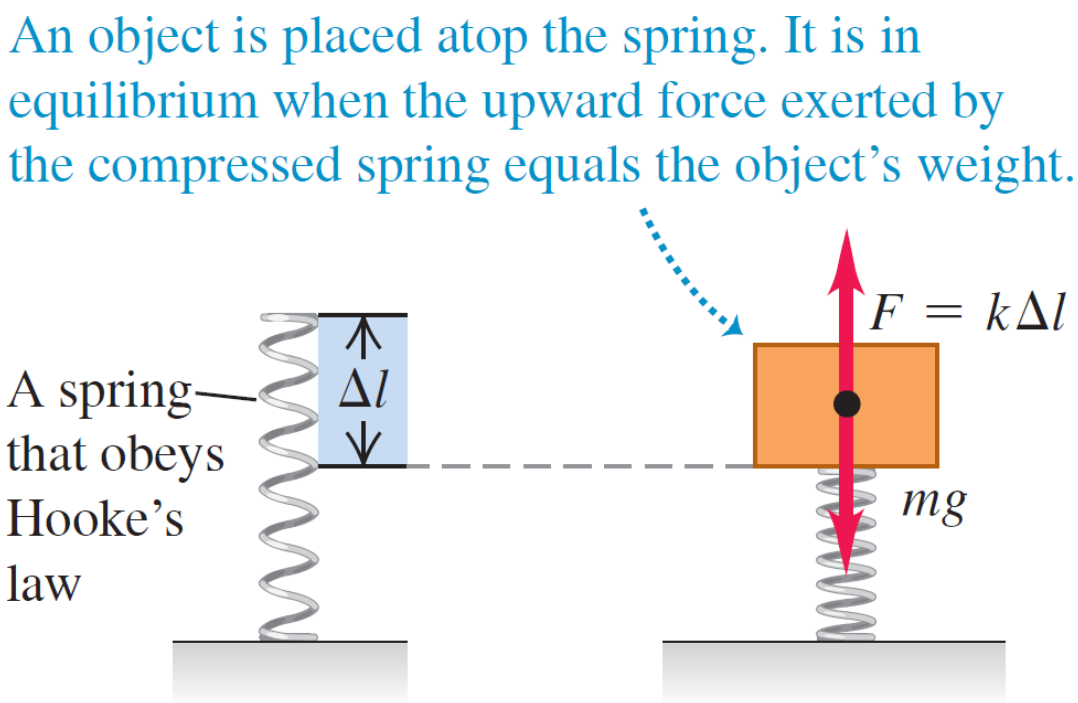
Vertical SHM (1 of 2)

- If an object oscillates vertically from a spring, the restoring force has magnitude *kx*. Therefore the vertical motion is SHM.



Vertical SHM (2 of 2)

- If the weight mg compresses the spring a distance Δl, the force constant is $k = \dfrac{mg}{\Delta l}$.

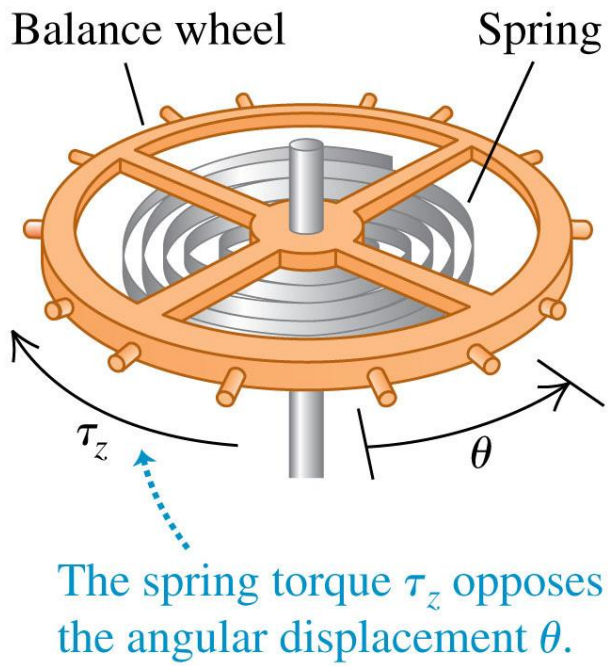


A mass-spring system in simple harmonic motion has an energy of 5.64 J and an amplitude of 13 cm. Find the maximum force that the spring exerts on the mass. (Be careful with units!)

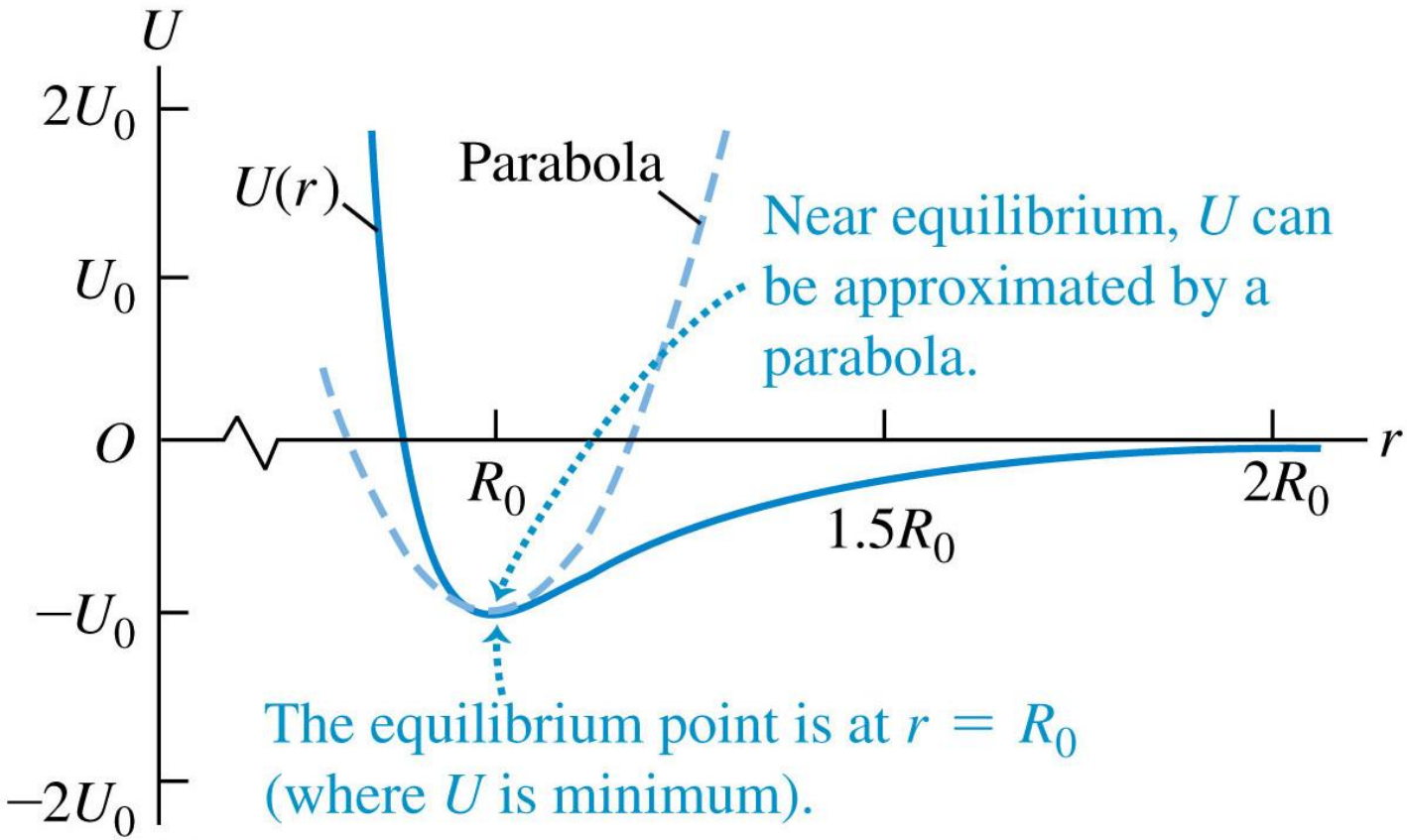
A simple harmonic oscillator consists of a block of mass 7 kg and a horizontal spring with constant 165 N/m. If the block is moved 0.34 m from equilibrium, stretching the spring, calculate the block's maximum acceleration a_{max}.

Angular SHM

- A coil spring exerts a restoring torque $\tau_z = -k\theta$, where k is called the **torsion constant** of the spring.
- The result is **angular** simple harmonic motion.



Potential Energy of a Two Atom System

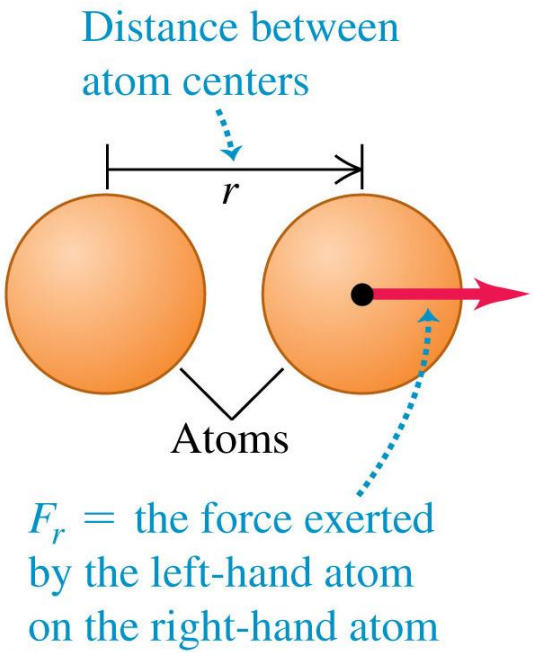


Vibrations of Molecules

- Shown are two atoms having centers a distance *r* apart, with the equilibrium point at $r = R_0$.
- If they are displaced a small distance *x* from equilibrium, the restoring force is approximately

$$F_r = -\left(\frac{72U_0}{R_0^2}\right)x$$

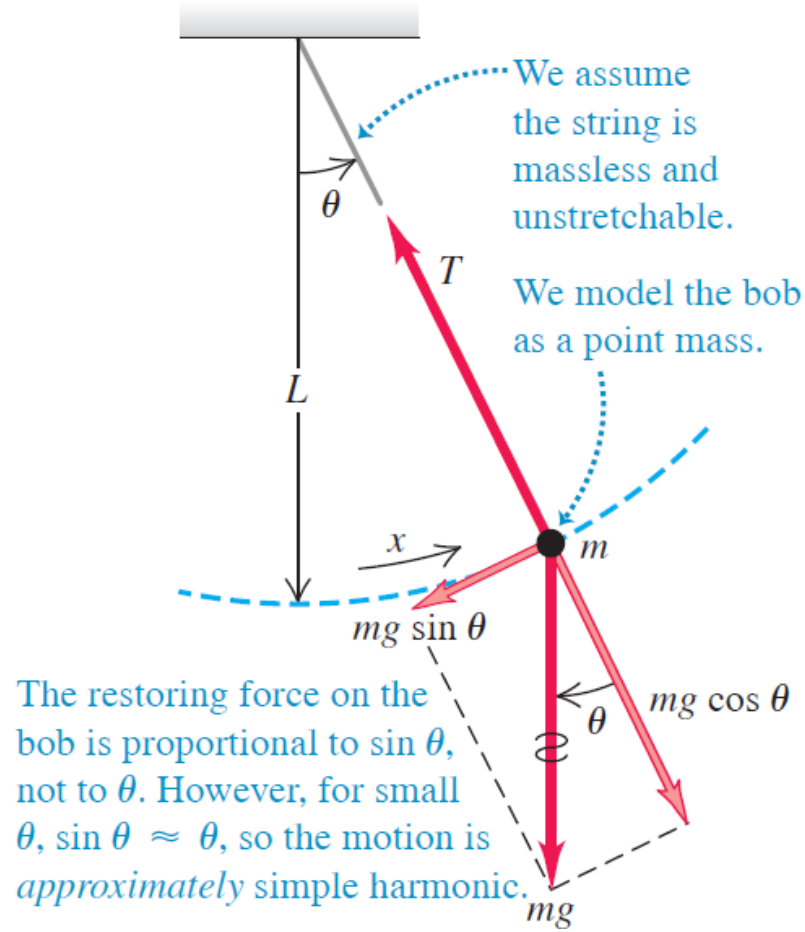
- So $k = \dfrac{72U_0}{R_0^2}$, and the motion is SHM.



The Simple Pendulum

- A **simple pendulum** consists of a point mass (the bob) suspended by a massless, unstretchable string.
- If the pendulum swings with a small amplitude θ with the vertical, its motion is simple harmonic.

(b) An idealized simple pendulum

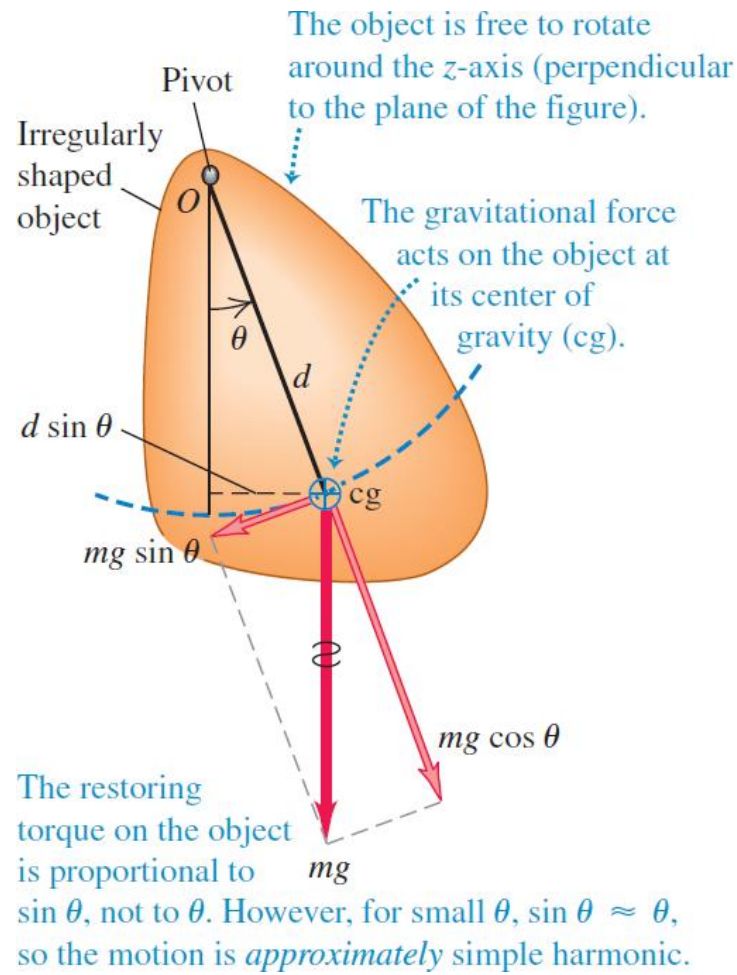


A simple pendulum has length 1.91 m and mass 8.6 kg. If the local acceleration of gravity on this planet is 6.1 m/s^2, calculate the period of the motion.

A simple pendulum has length 1.9 m, mass 1.9 kg. If the local acceleration of gravity on this planet is 7.98 m/s^2, calculate the frequency of the motion in Hz.

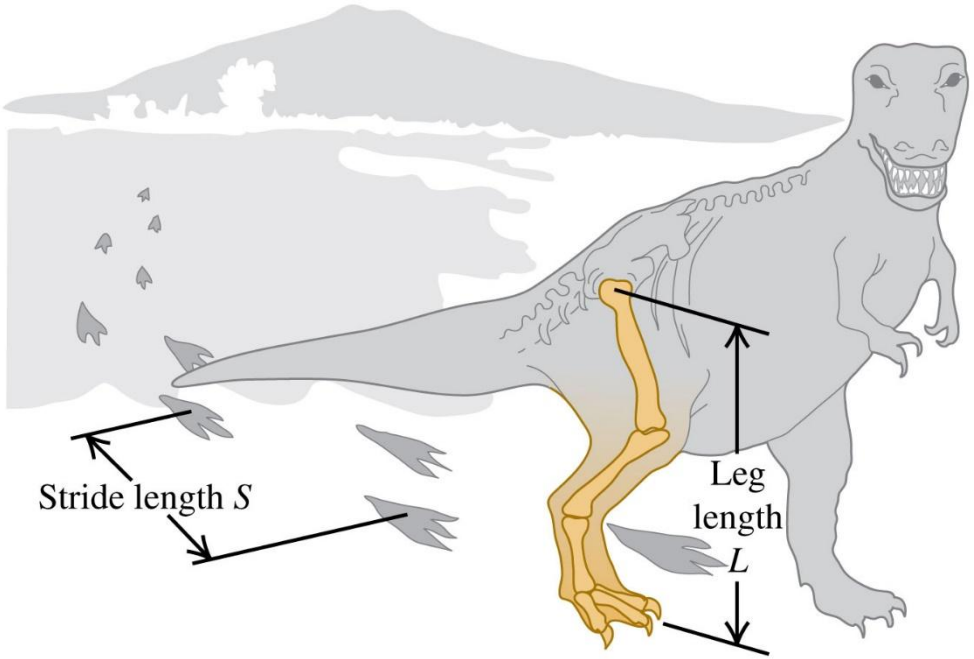
The Physical Pendulum

- A **physical pendulum** is any real pendulum that uses an extended object instead of a point-mass bob.
- For small amplitudes, its motion is simple harmonic.
- [Video Tutor Solution: Example 14.9]



Tyrannosaurus Rex and the Physical Pendulum

- We can model the leg of **Tyrannosaurus rex** as a physical pendulum.



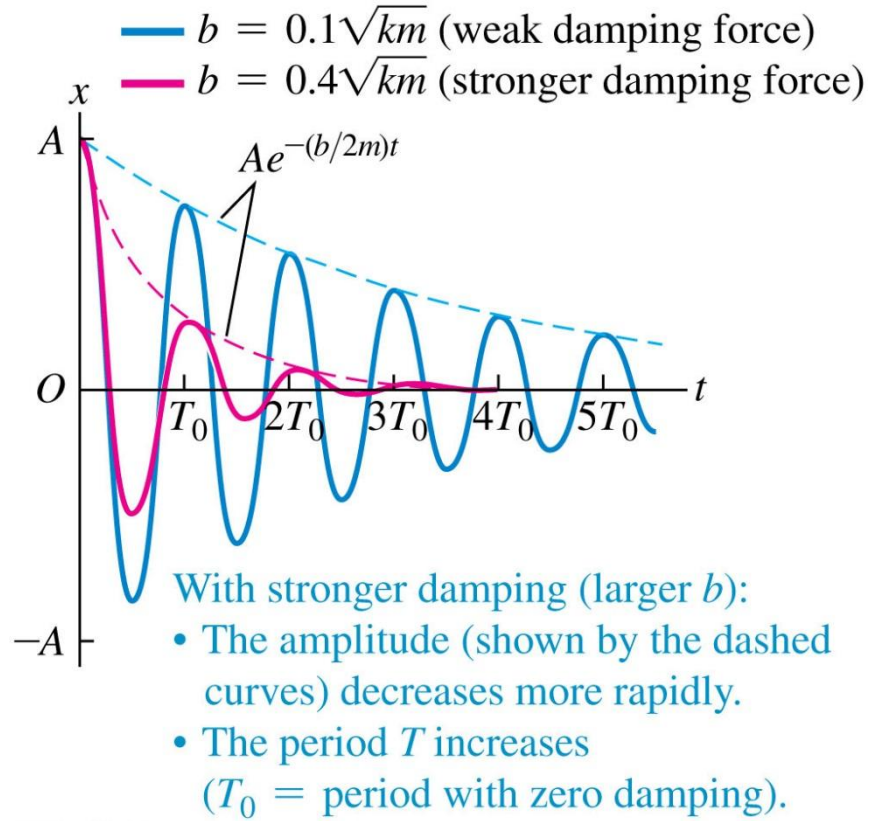
- Video Tutor Solution: Example 14.10

An object is free to oscillate around a pivot located a distance 0.88 m from its center of mass. Calculate the period of small oscillations if its mass is 5.6 kg and its moment of inertia about the pivot is 14.8 $kg \cdot m^2$.

An object is free to oscillate around a pivot that is a distance 0.45 m from its center of mass. Calculate the frequency of small oscillations if its mass is 5.1 kg and its moment of inertia is about the pivot is 1 $kg \cdot m^2$.

Damped Oscillations

- Real-world systems have some dissipative forces that decrease the amplitude.
- The decrease in amplitude is called **damping** and the motion is called **damped oscillation**.



Forced Oscillations and Resonance (1 of 2)

- A damped oscillator left to itself will eventually stop moving.
- But we can maintain a constant-amplitude oscillation by applying a force that varies with time in a periodic way.
- We call this additional force a **driving force**.
- If we apply a periodic driving force with angular frequency ω_d to a damped harmonic oscillator, the motion that results is called a forced oscillation or a driven oscillation.

Maximum value of driving force

Amplitude of a forced oscillator

$$A = \frac{F_{max}}{\sqrt{(k - m\omega_d^2)^2 + b^2\omega_d^2}}$$

Damping constant

Force constant of restoring force

Mass

Driving angular frequency

Forced Oscillations and Resonance (2 of 2)

- This lady beetle flies by means of a forced oscillation.
- Unlike the wings of birds, this insect's wings are extensions of its exoskeleton.
- Muscles attached to the inside of the exoskeleton apply a periodic driving force that deforms the exoskeleton rhythmically, causing the attached wings to beat up and down.
- The oscillation frequency of the wings and exoskeleton is the same as the frequency of the driving force.

Ex. 3 The tip of a tuning fork goes through 440 complete vibrations in 0.500 s. Find the angular frequency and the period of the motion.

Ex. 9 When a body of unknown mass is attached to an ideal spring with force constant 120 N/m, it is found to vibrate with a frequency of 6.00 Hz. Find a) the period of the motion; b) the angular frequency; c) the mass of the object.

Ex. 11 An object is undergoing SHM with a period of 0.900 s and amplitude 0.320 m. At $t = 0$ the object is at $x = 0.320$ m and is instantaneously at rest. Calculate the time it takes the object to go (a) from $x = 0.320$ m to $x = 0.160$ m and (b) from $x = 0.160$ m to $x = 0$.

Ex. 28 A harmonic oscillator has angular frequency ω and amplitude A. a) What are the magnitudes of the displacement and velocity when the elastic potential energy is equal to the kinetic energy? (Assume that $U = 0$ at equilibrium.) b) How often does this occur in each cycle? What is the time between occurrences? c) At an instant when the displacement is equal to $A/2$, what fraction of the total energy of the system is kinetic and what fraction is potential?